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EVEN END SEMESTER EXAMINATION

2025

Name of the Course: **B.tech**

Name of the Paper: **Machine Learning**

Semester: **6th**

Paper Code: **TCS 662**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q 1)

(2X10=20 Marks) (CO2, CO4, CO5)

- (a) Design a simple 2-layer feedforward neural network. Explain the role of the activation function and show forward pass computations with sample inputs and weights.?
- (b) Describe the Elbow Method and Silhouette Scores for determining the optimal number of clusters in K-means. Illustrate their application with a suitable example.
- (c) What is the concept of Maximum A Posteriori Estimation (MAP)? Compare and contrast it with Maximum Likelihood Estimation (MLE).

Q 2)

(2X10=20 Marks) (CO2, CO4, CO5, CO6)

- (a) Consider the data set shown in table below.

A	B	Class
1	0	Positive
0	0	Negative
0	1	Negative
0	1	Negative
0	0	Positive
1	0	Positive

Estimate all the parameters for training Naïve Bayes classifier and classify the given test sample (A=1, B=1) using estimated parameters.

- (b) Explain the Curse of Dimensionality. How does it affect the performance of machine learning models, especially clustering?
- (c) What are autoencoders? Describe their structure and provide an example of their real-world application.

Q 3) (2X10=20 Marks) (CO1, CO2)

- (a) Describe the architecture of a Convolutional Neural Network (CNN). What are the key components, and how does it differ from a fully connected network?
- (b) Discuss the concept of backpropagation in training neural networks. Provide an example of how weights are updated during the process.
- (c) What are the applications of deep learning and how is it useful in the real world?

Q 4) (2X10=20 Marks) (CO2, CO5)

- (a) Explain the difference between K-Means and DBSCAN clustering techniques. Give an example of a situation where DBSCAN would work better than K-Means.
- (b) Apply the K-means algorithm (one iteration only) on the following data points with $k=2$: Points: (1, 2), (1, 4), (5, 6), (6, 5). Initial centroids: (1, 2) and (5, 6).
- (c) Provide an overview of reinforcement learning. How does it differ from supervised and unsupervised learning paradigms?

Q 5) (2X10=20 Marks) (CO4, CO6)

- a) Explain the logistic regression model. Derive the loss function using cross-entropy.
- b) Define a perceptron. Explain how the perceptron algorithm works with a simple example.
- c) Explore activation functions in neural networks. Provide examples of one non-linear activation function and one linear activation function, and discuss their significance.

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